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RICE LEAF DISEASE CLASSIFICATION
DeiT-Small | Canonical V1 Dataset
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CONFIGURATION
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Device          : cuda
Model           : deit_small_patch16_224
Image size      : 224 x 224
Batch size      : 32
Epochs         : 25
Learning rate   : 0.0001
Weight decay    : 0.01
Seed           : 42
Num classes     : 6
Workers        : 2
Mixed precision : True
GPU            : Tesla T4
GPU memory      : 14.56 GB

Dataset root    : /kaggle/input/datasets/varun2ks05/newdata
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VERIFYING CANONICAL DATASET
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Canonical dataset found successfully.

Dataset structure:

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Train : /kaggle/input/datasets/varun2ks05/newdata/train
Val   : /kaggle/input/datasets/varun2ks05/newdata/val
Test  : /kaggle/input/datasets/varun2ks05/newdata/test
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Canonical class mapping loaded.

Class mapping:

```
0 -> Bacterial Leaf Blight
1 -> Brown Spot
2 -> Healthy Rice Leaf
3 -> Leaf Blast
4 -> Leaf scald
5 -> Sheath Blight
```

Checking class directories...

Train : 6 classes verified
Validation : 6 classes verified
Test : 6 classes verified

LOADING CANONICAL DATASET

ImageFolder class mapping:

{'Bacterial Leaf Blight': 0, 'Brown Spot': 1, 'Healthy Rice Leaf': 2,
'Leaf Blast': 3, 'Leaf scald': 4, 'Sheath Blight': 5}

Class mapping verified successfully.

Dataset sizes:

Training : 3063
Validation : 383
Test : 383

TEST SET STATUS: LOCKED / NOT USED

CANONICAL DATASET DISTRIBUTION

Class	Train	Validation	Test
Bacterial Leaf Blight	509	63	64
Brown Spot	517	65	64
Healthy Rice Leaf	522	66	65
Leaf Blast	507	63	64
Leaf scald	502	63	63
Sheath Blight	506	63	63

INITIALIZING DeiT-SMALL

Model initialized: deit_small_patch16_224

STARTING DeiT TRAINING

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Model : deit_small_patch16_224
Epochs : 25
Train : 3063 images
Validation : 383 images
Batch size : 32
Device : cuda

Epoch [01/25] | Train Loss: 0.6593 Acc: 76.27% | Val Loss: 0.3802 Acc:
86.68% | LR: 9.96e-05 | Time: 36.1s
--> NEW BEST MODEL SAVED (86.68%)

Epoch [02/25] | Train Loss: 0.2587 Acc: 90.79% | Val Loss: 0.1842 Acc:
94.26% | LR: 9.84e-05 | Time: 27.5s
--> NEW BEST MODEL SAVED (94.26%)

Epoch [03/25] | Train Loss: 0.1461 Acc: 95.17% | Val Loss: 0.1364 Acc:
95.30% | LR: 9.65e-05 | Time: 27.4s
--> NEW BEST MODEL SAVED (95.30%)

Epoch [04/25] | Train Loss: 0.0961 Acc: 97.26% | Val Loss: 0.0766 Acc:
98.17% | LR: 9.39e-05 | Time: 27.7s
--> NEW BEST MODEL SAVED (98.17%)

Epoch [05/25] | Train Loss: 0.0669 Acc: 97.91% | Val Loss: 0.1297 Acc:
95.82% | LR: 9.05e-05 | Time: 27.8s

Epoch [06/25] | Train Loss: 0.0555 Acc: 98.33% | Val Loss: 0.1292 Acc:
96.08% | LR: 8.66e-05 | Time: 28.0s

Epoch [07/25] | Train Loss: 0.0412 Acc: 98.69% | Val Loss: 0.0416 Acc:
98.69% | LR: 8.21e-05 | Time: 27.8s
--> NEW BEST MODEL SAVED (98.69%)

Epoch [08/25] | Train Loss: 0.0380 Acc: 98.89% | Val Loss: 0.1113 Acc:
95.82% | LR: 7.70e-05 | Time: 28.0s

Epoch [09/25] | Train Loss: 0.0350 Acc: 99.05% | Val Loss: 0.0623 Acc:
97.39% | LR: 7.16e-05 | Time: 27.3s

Epoch [10/25] | Train Loss: 0.0205 Acc: 99.38% | Val Loss: 0.0868 Acc:
96.61% | LR: 6.58e-05 | Time: 27.8s

Epoch [11/25] | Train Loss: 0.0282 Acc: 99.09% | Val Loss: 0.0409 Acc:
98.43% | LR: 5.98e-05 | Time: 27.9s

Epoch [12/25] | Train Loss: 0.0159 Acc: 99.54% | Val Loss: 0.0905 Acc:
97.39% | LR: 5.36e-05 | Time: 26.9s

Epoch [13/25] | Train Loss: 0.0067 Acc: 99.80% | Val Loss: 0.0587 Acc:
98.17% | LR: 4.74e-05 | Time: 27.5s

Epoch [14/25] | Train Loss: 0.0097 Acc: 99.74% | Val Loss: 0.0324 Acc:
98.96% | LR: 4.12e-05 | Time: 27.1s

--> NEW BEST MODEL SAVED (98.96%)

Epoch [15/25] | Train Loss: 0.0059 Acc: 99.87% | Val Loss: 0.0397 Acc:
98.69% | LR: 3.52e-05 | Time: 28.5s

Epoch [16/25] | Train Loss: 0.0024 Acc: 99.97% | Val Loss: 0.0392 Acc:
98.96% | LR: 2.94e-05 | Time: 27.2s

Epoch [17/25] | Train Loss: 0.0035 Acc: 99.90% | Val Loss: 0.0306 Acc:
99.22% | LR: 2.40e-05 | Time: 28.4s

--> NEW BEST MODEL SAVED (99.22%)

Epoch [18/25] | Train Loss: 0.0071 Acc: 99.67% | Val Loss: 0.0589 Acc:
98.17% | LR: 1.89e-05 | Time: 26.9s

Epoch [19/25] | Train Loss: 0.0035 Acc: 99.90% | Val Loss: 0.0274 Acc:
98.96% | LR: 1.44e-05 | Time: 28.3s

Epoch [20/25] | Train Loss: 0.0048 Acc: 99.90% | Val Loss: 0.0474 Acc:
98.17% | LR: 1.05e-05 | Time: 28.1s

Epoch [21/25] | Train Loss: 0.0013 Acc: 100.00% | Val Loss: 0.0400 Acc:
98.17% | LR: 7.12e-06 | Time: 27.6s

Epoch [22/25] | Train Loss: 0.0022 Acc: 99.93% | Val Loss: 0.0341 Acc:
98.69% | LR: 4.48e-06 | Time: 27.1s

Epoch [23/25] | Train Loss: 0.0011 Acc: 100.00% | Val Loss: 0.0355 Acc:
98.69% | LR: 2.56e-06 | Time: 27.5s

Epoch [24/25] | Train Loss: 0.0013 Acc: 100.00% | Val Loss: 0.0352 Acc:
98.69% | LR: 1.39e-06 | Time: 27.8s

Epoch [25/25] | Train Loss: 0.0012 Acc: 100.00% | Val Loss: 0.0356 Acc:
98.69% | LR: 1.00e-06 | Time: 27.5s

TRAINING COMPLETED

Total training time : 11m 43s

Best validation accuracy : 99.22%

Best epoch : 17

Best model :

/kaggle/working/DeiT_Canonical_V1/models/best_deit_small_canonical_v1.pth

LOADING BEST MODEL FOR FINAL VALIDATION

Loaded best checkpoint from epoch 17

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FINAL VALIDATION METRICS
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Overall Accuracy : 99.22%
Macro Precision  : 99.22%
Macro Recall     : 99.21%
Macro F1-Score   : 99.21%
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DETAILED CLASSIFICATION REPORT
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	precision	recall	f1-score	support
Bacterial Leaf Blight	1.0000	0.9683	0.9839	63
Brown Spot	1.0000	1.0000	1.0000	65
Healthy Rice Leaf	1.0000	1.0000	1.0000	66
Leaf Blast	0.9844	1.0000	0.9921	63
Leaf scald	0.9688	0.9841	0.9764	63
Sheath Blight	1.0000	1.0000	1.0000	63
accuracy			0.9922	383
macro avg	0.9922	0.9921	0.9921	383
weighted avg	0.9923	0.9922	0.9922	383

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CONFUSION MATRIX
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[[61  0  0  0  2  0]
 [ 0 65  0  0  0  0]
 [ 0  0 66  0  0  0]
 [ 0  0  0 63  0  0]
 [ 0  0  0  1 62  0]
 [ 0  0  0  0  0 63]]
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CLASS-WISE ACCURACY
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Bacterial Leaf Blight : 96.83%
Brown Spot : 100.00%
Healthy Rice Leaf : 100.00%
Leaf Blast : 100.00%
Leaf scald : 98.41%
Sheath Blight : 100.00%

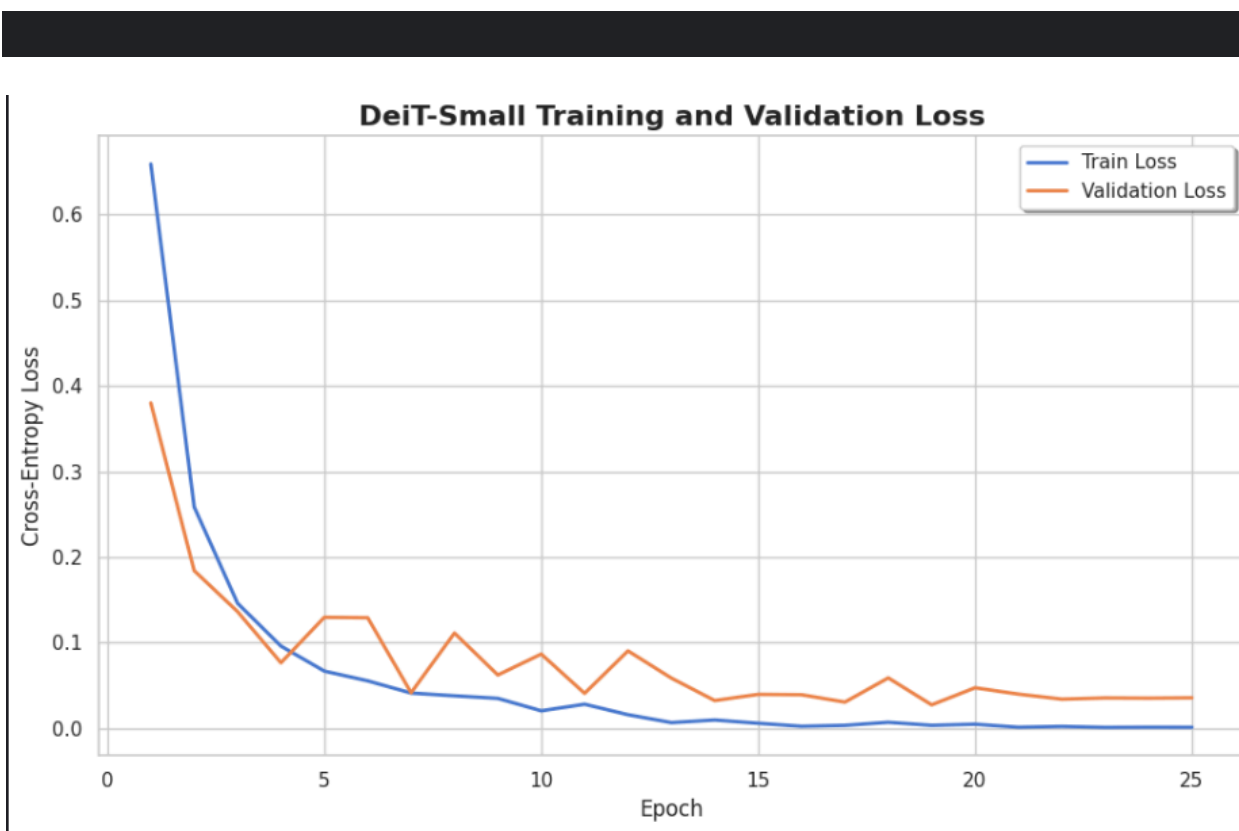
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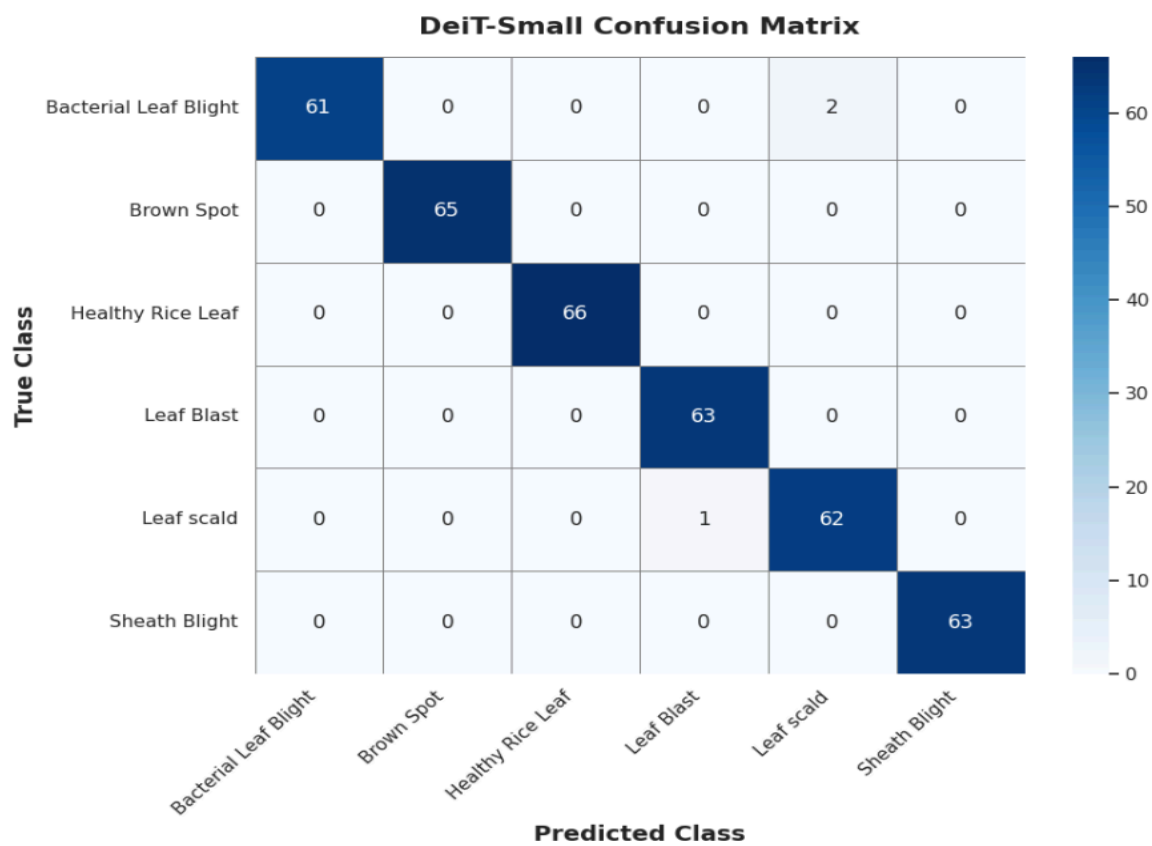
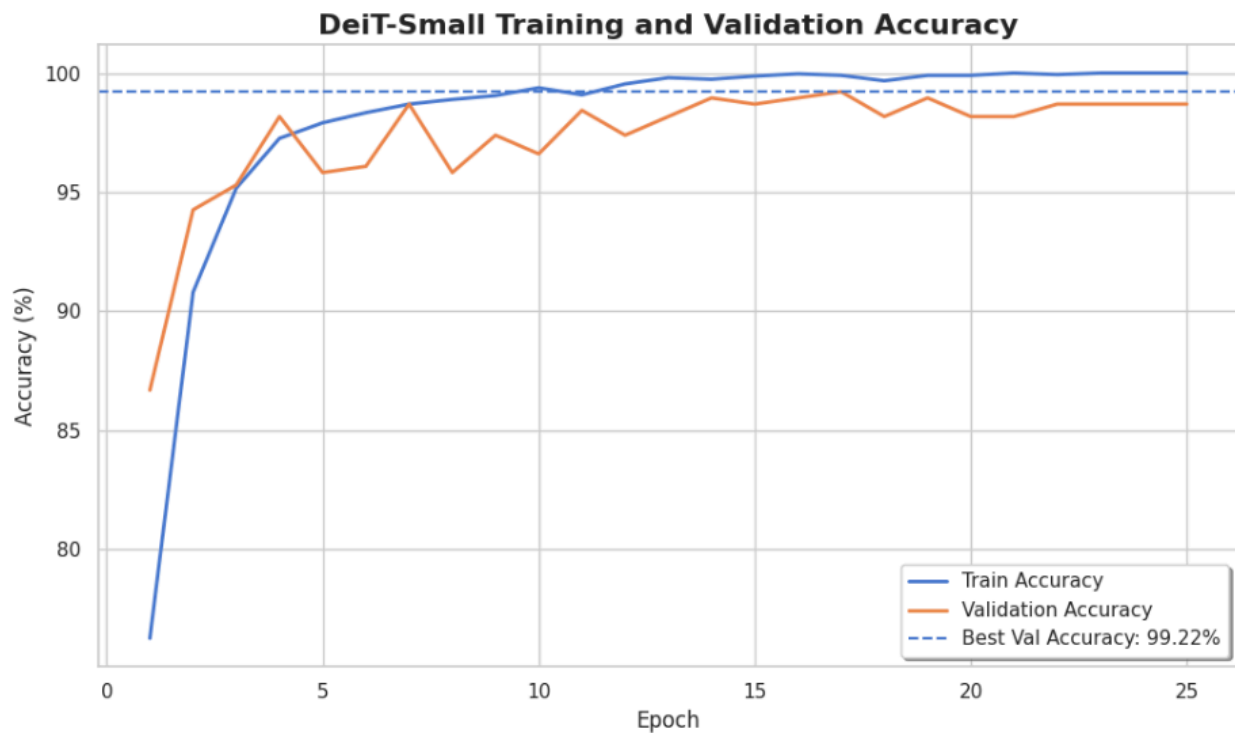
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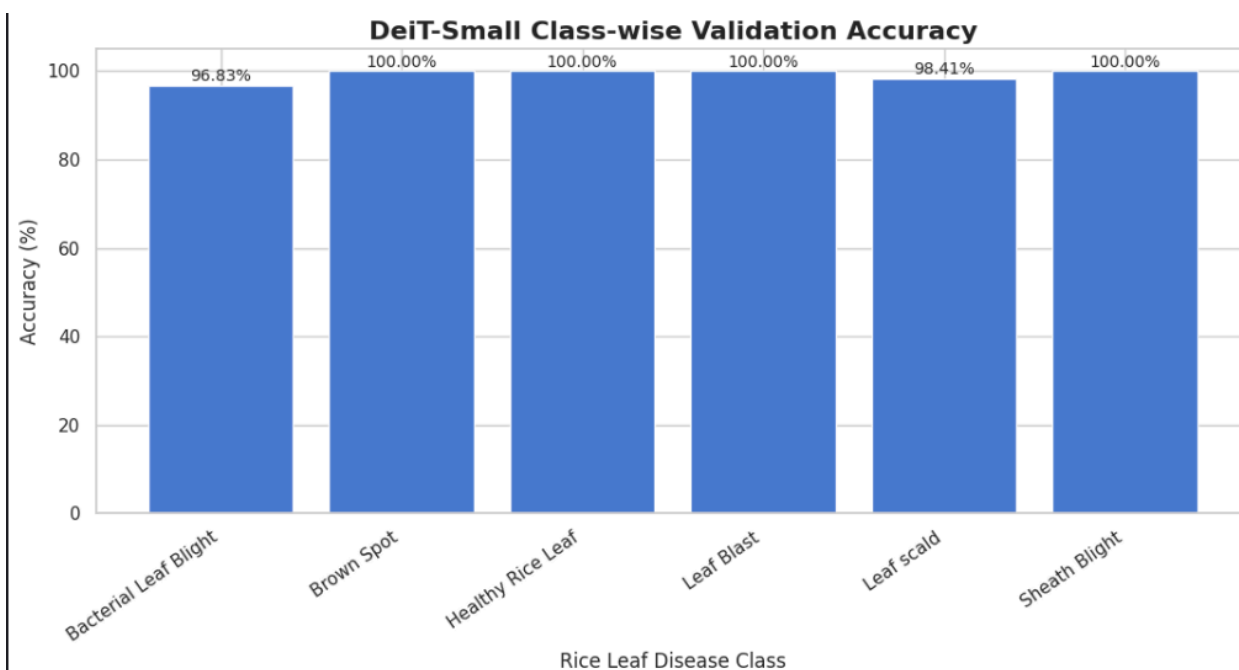
GENERATING TRAINING CURVES

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EXPERIMENT COMPLETE

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BEST MODEL

/kaggle/working/DeiT_Canonical_V1/models/best_deit_small_canonical_v1.pth

TRAINING HISTORY

/kaggle/working/DeiT_Canonical_V1/training_history.csv

METRICS

/kaggle/working/DeiT_Canonical_V1/final_metrics.json

PLOTS

/kaggle/working/DeiT_Canonical_V1/plots/deit_training_validation_loss.png
/kaggle/working/DeiT_Canonical_V1/plots/deit_training_validation_accuracy.
png
/kaggle/working/DeiT_Canonical_V1/plots/deit_confusion_matrix.png

/kaggle/working/DeiT_Canonical_V1/plots/deit_class_wise_accuracy.png

CLASSIFICATION REPORT

/kaggle/working/DeiT_Canonical_V1/reports/classification_report.txt

PDF REPORT

/kaggle/working/DeiT_Canonical_V1/DeiT_Canonical_V1_Experiment_Report.pdf

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FINAL VALIDATION RESULT

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Accuracy	: 99.22%
Precision	: 99.22%
Recall	: 99.21%
F1 Score	: 99.21%
Best Epoch	: 17
Best Val Acc	: 99.22%

TEST SET: NOT EVALUATED / LOCKED

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DONE

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